



TEMPLATE-DESIGNED ORGANIC ELECTRONICS
RESEARCH TRAINING GROUP 2591

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Influence of adsorbate-adsorbate interactions and template effects on optical properties of organic films

TIDE project ID: C1
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1 State of the art

Optical properties of thin organic films of π -conjugated molecules are most important for many modern applications such as organic solar cells (OSCs)^[1] or nonlinear optics^[2]. Although, the optical properties of a monomer are important for an organic film, these get altered by intermolecular coupling of the transition dipole moments during structure formation.^[3]

In molecular crystals, neighbored molecules can interact with each other by electromagnetic interactions. In dependence of the transition dipole moment, the intermolecular distance, and the molecular packing, an excitation gets delocalized across the aggregate. This results in a shift of the absorption band to shorter (H-aggregates) or longer wavelengths (J-aggregates). While for H-aggregates the fluorescence yield is in most cases lower than for the respective monomer, J-aggregates show high fluorescence yields with narrow band widths and small Stokes shifts.^[4] For some years, these shifts in the absorption bands of π -conjugated molecules were intensively investigated for molecular crystals and thick films in order to improve device efficiencies.^[5]

Thus, the optical properties of an aggregate can be manipulated by adjusting the molecular packing.^[1] Differing from the situation in the bulk crystal, structure formation in thin molecular films, especially in the monolayer (ML), is not only affected by intermolecular interactions but also depends strongly on the interactions of the molecules with the underlying template.^[6,7,8] Depending on the binding strength of the molecules to a template (i.e. the strength of the physisorptive or chemisorptive bonding), the ratio between the intermolecular interactions and the molecule-template interactions change with variation of the template. Examples for surfaces with different reactivities are Cu(100), Ag(100), Au(100), and KCl(100). Please note that, in principle, all of these fcc(100) surfaces show the same symmetry, but the Au(100) surface shows a quasihexagonal reconstruction of its surface.^[9] This leads to the second structure determining aspect of a template, which is surface symmetry. Considering different crystal structures of the template (e.g. fcc and bcc) and the symmetry of different surfaces (e.g. (100), (100), (111)), different adsorption sites and symmetries are provided by the template. These differing arrangement of adsorption sites is also strongly affecting the monolayer structure of an adsorbate.

A group of molecules that are well known for the formation of H- and J-aggregates are merocyanines.^[1,5] They are build up by a heterocyclic electron donating and accepting part connected via an polymethine bridge consisting of an odd number of methine groups $(CH)_n$.^[3] A merocyanine with quite high electrostatic dipole moment ($\mu_g=13-14$ D) is 2-[5-(5-dibutylamino-thiophen-2-yl-methylene)-4-*tert*-butyl-5*H*-thiazol-2-ylidene]-malononitrile (in the following abbreviated as HB238).^[1] Due to this property, HB238 interacts strongly with neighboring molecules and tends to form dimers in the crystalline state.^[10] Although, this structure formation is already well studied,^[10] only small information about the self-organization of HB238 and its derivatives on a surface acting as a template are known.^[11]

The aim of the present project is to study the self-organization of different merocyanines, on three metal fcc(100) templates in order to learn more about the influence of the template on the intermolecular structures in the merocyanine layers. Finally, an KCl(100) surface will be used as a template to study the self-organization on an inert wide-band-gap surface and furthermore the resulting optical properties of the organic layer.

2 Research vision & project idea

The vision of this project is to develop knowledge how the optical properties of an organic film can be modified by changing the template or adjusting the intra-molecular structure. To reach this goal this project is divided into three different parts which will be explained in more detail in the following paragraphs.

Part 1: The first part investigates the self-organization of a series of merocyanines on a Ag(100) surface. The Ag(100) surface was chosen as a starting template because of its medium reactivity, to avoid a dissociative adsorption. In addition, Ag(100) is an unreconstructed surface. The latter one in combination with the four-fold symmetry of the (100) surface reduces the number of symmetrical equivalent domains formed by the merocyanines and therefore facilitates the structure determination.

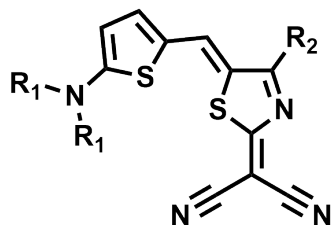


Figure 1: Basic molecular structure of the merocyanine derivatives. Groups $\mathbf{R}_1$ and $\mathbf{R}_2$ are depicted in Table 1.

Table 1: Overview about the structural differences of the merocyanines in $\mathbf{R}_1$ and $\mathbf{R}_2$.

Molecule	$\mathbf{R}_1$	$\mathbf{R}_2$
HB238 (A)	<i>n</i> Bu	<i>t</i> Bu
Derivative B	<i>n</i> Pr	<i>t</i> Bu
Derivative C	Et	<i>t</i> Bu
Derivative D	Me	<i>t</i> Bu
Derivative E	<i>n</i> Bu	<i>i</i> Pr

To keep the changes in the self-organization of the merocyanine molecules comparable, we decided to study a series of merocyanine derivatives which do only deviate from the starting molecule HB238 in one specific group. Starting with HB238 ($\mathbf{R}_1 = n\text{Bu}$) we reduce the length of the alkyl chains acting as donor substituents ($\mathbf{R}_1 = n\text{Pr}, \text{Et}, \text{Me}$) such that we obtain a series of four derivatives. From first results (see chapter 3), we derive that these alkyl groups show weak bonding to the Ag(100) surface (template) but act as spacers in the molecular arrangement within the unit cell. Hence, the bonding strength of the molecules to the template should not change within the series of these derivatives, and possible structural changes are expected as a result of different space requirement, only.

In addition to this series, an other derivative is investigated which molecular structure differs from HB238 in the acceptor substituent. As HB238 tends to dewet from the Ag(100) at room temperature (RT) we plan to change the *tert*-butyl group (*t*Bu) against an *iso*-propyl group (*i*Pr) which is expected to yield a stronger bonding of the molecule to the template and hence a stronger bonding of the π -system. This stronger bonding of the π -system is motivated by a reduced bonding distance of the molecule to the surface which was before terminated by the sterical requirement of the *tert*-butyl group. This slightly stronger bonding of the derivative to the template could cause a better stability of the organic film, avoiding a detwetting, and reducing the tendency of the molecules to undergo cluster formation.

Part 2: In part two of the project, the self-organized structures of the different derivatives are determined on Cu(100), Au(100)-hex, and finally on a KCl(100) layer epitaxially grown on Ag(100). This is planned in order to study the structural changes that are induced by chemically different templates.

Cu and Au surfaces are in comparison to the Ag surface of higher and lower reactivity. For Cu(100) the symmetry of the surface is the same as for Ag(100), but due to the smaller lattice constant and the higher reactivity, changes in the self-organization of the molecules may be expected. Au is less reactive than Ag and the respective bulk lattice constants are quite similar, but the situation for the Au(100) surface is more complex. The Au(100) surface is less reactive than the Ag surface and undergoes a quasi-hexagonal reconstruction of the topmost layer, wherefore it is denoted as Au(100)-hex.^[9] Due to this reconstruction the surface does not remain flat, but exhibits a nanowire-like topology.^[9] This leads to the question, whether merocyanines are able to bind to the reconstructed Au(100) surface and whether a binding causes a lifting of the reconstruction.

In addition, the merocyanines will be investigated on a KCl(100) layer which is epitaxially grown on the Ag(100) template. KCl is an insulator with a band gap of 8.7 eV.^[12] This property is required to decouple the organic film from the underlying metal surface in order to avoid a quenching of the fluorescence of an organic molecule on this film. On the other hand, bonding of the molecules to the insulating layer might be rather weak and intermolecular interactions may be suspected to dominate the self-organization. Hence, a template, i.e. interface controlled self-organization of merocyanines on KCl(100) might be difficult. However, it is most important for part three of the project, namely the investigation of fluorescence of the organic layers. Therefore, based on the results of the measurements, we consider the possibility to change the template in this step and to choose a different insulator material.

Part 3: The third part of the project includes the measurements of laser induced fluorescence of the organic layers on KCl(100) or an equivalent insulator surface. Applying the knowledge obtained in part one and two, we plan a modification of the observed fluorescence by adjustment of the respective molecular structures of the organic films. Due to the expected weak bonding of the merocyanines to insulator surfaces, the intermolecular interactions will dominate and hence, this modification will mainly be performed by using the above described derivatives of HB238. This will in particular allow to quantify the changes observed in the fluorescence in dependence of the length of the donor substituting alkyl chains.

Applied techniques: All organic films will be investigated by spot profile low energy electron diffraction (SPA-LEED) in order to determine the unit cell dimensions of the superstructures and to identify temperature dependent phase transitions in the organic films. These measurements are supplemented by scanning tunneling microscopy (STM) measurements which allow a local quantification of the domain size distributions and a predication about the unit cell content. To gain more informations about the binding situation of the molecules to the surface and to make sure that the molecules do not dissociate, additional X-ray photoelectron spectroscopy (XPS) measurements are performed by the research group of Dr. S. Olthof. The proposed structure models for the organic monolayers are additionally verified and optimized by density functional theory (DFT) performed in the group of Prof. Dr. T. Bredow. Finally, the optical properties of the organic films are investigated by laser induced fluorescence. All measurements will be performed under ultra high vacuum (UHV) at base pressures of about 3×10^{-10} mbar.

Context in the TIDE project: The investigation of template effects of substrates is one of the main goals of the TIDE project. The above described research project strongly supports this goal by describing structural changes in dependency of the chosen templates and, furthermore, by using this knowledge to improve optical properties of organic films. Furthermore, this project is set up as a cooperation of three TIDE focus areas, namely B1 (provided molecules), C3 (XPS measurements), and E2 (DFT calculations), supporting and strengthening the TIDE research concept.

3 Preliminary work and first results

The above described project started with the investigation of HB238 on a Ag(100) template. To the best of our knowledge, we prepared the first ordered thin films (1–2 ML) of a merocyanine on a metal surface. Because of the novelty of this adsorbate system a number of different complementary methods were used in order to characterize the adsorption behavior of merocyanines on Ag(100), namely, SPA-LEED, STM, and XPS/UPS. It turned out that vacuum deposition of thin HB238 films is possible and that the molecule does not dissociate on the Ag(100) surface.^[13,14] Furthermore, we found the formation of two different phases, α and β .^[13]

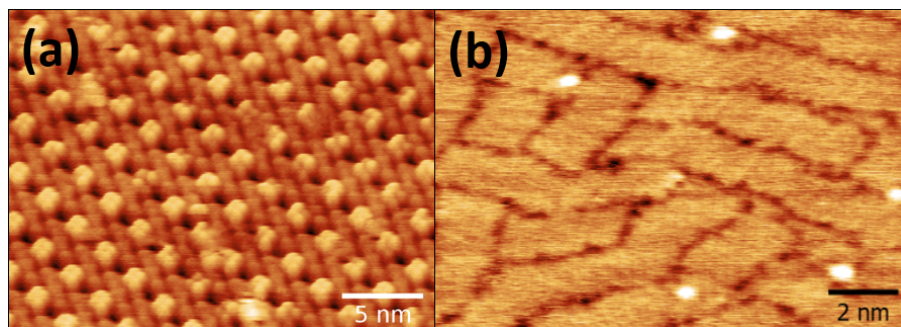


Figure 2: STM images of HB238 on Ag(100). (a) α -phase ($U_{\text{bias}} = -0.75$ V; $I_t = 4.8$ pA); (b) β -phase ($U_{\text{bias}} = -0.51$ V; $I_t = 9$ pA).

Figure 2 shows STM images of the two phases. While the α -phase [Figure 2 (a)] was found to be commensurate, which possibly indicates a chemisorptive binding of the π -system to the template, this is different for the β -phase [Figure 2 (b)]. There, the intramolecular interactions dominate, leading to a chain formation of HB238 molecules and presumably weaker physisorptive bonding to the Ag(100) surface. To explain the differences in the two phases we developed first structure models, which are presently still under investigation by further STM experiments and DFT calculations.

After modifications and repairs on our UHV chambers, we spent much effort to verify the above mentioned data and to improve the quality of the STM measurements e.g. by STM-tip modification and noise reduction. In addition we recently performed bias dependent STM measurements to gain additional data to check the validity of our structure model.^[15] In a last step, we started with the preparations of the measurements we planned to perform for the series of derivatives of HB238. We are prepared to perform high quality STM measurements for the series of derivatives, starting with derivative D (see Figure 1 and Table 1).

4 Project milestone & working plan for the next 6 months

As already described in Chapter 2 we divided the project into three different parts. To have enough time to perform all experiments with adequate accuracy, we plan to have the final results of part 3 at the end of quarter 2-2023. In this way two quarters (3 and 4) in 2023 remain for writing the PhD-thesis. Table 3 shows the complete time schedule for part 1 to part 3 of our project. In the following paragraphs we will introduce each time slot shown in Table 3. For further leading information please see Chapter 2. Please note that due to experimental results and difficulties this time schedule is preliminary and might change during the project.

Figure 3: Preliminary time schedule for the experimental work of the above described project.

Year	Quarter 1	Quarter 2	Quarter 3	Quarter 4
2021	Reparations and Modifications on UHV chambers		Part 1 (1. milestone)	
2022	Part 1 (2. milestone)	Part 2 (3.-5. milestone)		
2023	Part 3 (6. milestone)		Writing of PhD-thesis	

After we have successfully repaired and modified our UHV chambers (quarter 1 and 2 of 2021), we will start with part 1 of our project. Part 1 includes the verification of our structure models for HB238 (**first milestone**) and the structure determination of the derivatives on Ag(100) (**second milestone**). Due to the large number of four derivatives (see Figure 1 and Table 1) we plan to investigate these structures in three quarters (3,4, and 1 of 2021/22). Part 1 ends with achievement of the **second milestone**, namely the determination of structure models for the four derivatives on Ag(100).

After we have successfully determined the structures of HB238 and its derivatives on Ag(100), part 2 of our project will start around the 2. quarter of 2022. In this part we introduce HB238 to two further metal surfaces (Cu(100) and Au(100)-hex) and KCl(100). Due to the high importance of the structure formation of HB238 (and optionally also its derivatives) on KCl(100) for all following optical experiments, these structures will be thoroughly investigated to ensure reproducibility and a good understanding of the structure formation in the monolayer. Due to the three different new templates, part 2 of our project includes in total three milestones. The (**third milestone**) and (**fourth milestone**) of our project are the determined structures of HB238 on Cu(100) and Au(100)-hex, respectively. The (**fifth milestone**), one of the most important, will be the achieved structure formation of HB238 (and optionally also of its derivatives) on KCl(100), the requirement for part 3 of our project.

The final part of our project, part 3 focuses on the optical measurements of HB238 (and optionally also its derivatives) on KCl(100). Following the above time schedule (Table 3), these measurements will be performed in the 1. and 2. quarter of 2023. Assuming the experiments to be successful, this part of the project will end with the achievement of the (**sixth milestone**), namely the determination of the optical differences of the monolayer compared to the one of the monomer and the bulk structures of the molecules. Furthermore, depending on the remaining time, also measurements with the derivatives of HB238 will be performed to document changes in the optical properties due to the different intramolecular structures.

Research residence: Due to the actual Covid19 situation, the exact time slot for the three-month research residence cannot be scheduled. Based on the working plan presented above, we would prefer to start the research residence in the first to second quarter of 2022. To complement the above named measuring techniques and results, we favor the research group Peter Grutter at the department of Physics of the McGill University, who performs high quality AFM measurements.

References

- [1] A. Liess, A. Arjona-Esteban, A. Kudzus, J. Albert, A.-M. Krause, A. Lv, M. Stolte, K. Meerholz, and F. Würthner. Ultranarrow Bandwidth Organic Photodiodes by Exchange Narrowing in Merocyanine H- and J-Aggregate Excitonic Systems. *Adv. Funct. Mater.* **29**, 1805058 (2019).
- [2] H. S. Nalwa and S. Miyata (ed.) *Nonlinear Optics of Organic Molecules and Polymers* (CRC Press, 1997).
- [3] F. Würthner, T. E. Kaiser, and C. R. Saha-Möller. J-aggregates: From serendipitous discovery to supramolecular engineering of functional dye materials. *Angewandte Chemie International Edition* **50**, 3376–3410 (2011).
- [4] T. Kobayashi (ed.) *J-Aggregates* (World Scientific, 1996).
- [5] A. Arjona-Esteban *et al.* Influence of solid-state packing of dipolar merocyanine dyes on transistor and solar cell performances. *Journal of the American Chemical Society* **137**, 13524–13534 (2015).
- [6] O. Bauer *et al.* Role of functional groups in surface bonding of planar π -conjugated molecules. *Phys. Rev. B* **86**, 235431 (2012).
- [7] S. Duhm, A. Gerlach, I. Salzmann, B. Bröker, R. L. Johnson, F. Schreiber, and N. Koch. PTCDA on Au(111), Ag(111) and Cu(111): Correlation of interface charge transfer to bonding distance. *Organic Electronics* **9**, 111–118 (2008).
- [8] D. P. Goronzy *et al.* Supramolecular assemblies on surfaces: Nanopatterning, functionality, and reactivity. *ACS Nano* **12**, 7445–7481 (2018).
- [9] A. Trembulowicz, B. Pieczyrak, L. Jurczyszyn, and Grażyna Antczak. Coexistence of nanowire-like hex and (1×1) phases in the topmost layer of au(100) surface. *Nanotechnology* **30**, 045704 (2018).
- [10] Private communication with K. Meerholz and D. Hertel (2019).
- [11] T. Limböck, D. Hertel, and K. Meerholz. Investigation and tuning of merocyanine dye thin films on ultra-flat substrates (2016). Poster contribution to Euro AFM Forum.
- [12] N. O. Lipari and A. B. Kunz. Energy bands for kcl. *Phys. Rev. B* **4**, 4639–4640 (1971).
- [13] A. J. Kny. *Adsorption and Ordering of the Merocyanine HB238 on the Ag(100) Surface*. Master’s thesis, Universität Bonn (2020).
- [14] Private communication with S. Olthof and M. Reimer (2020/2021).
- [15] N. Al-Shamery. *STM-investigation of the merocyanine HB238 and its methyl-derivative on the Ag(100) surface at varying Bias-voltage*. Bachelor’s thesis, Universität Bonn (2021). In preparation.